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Project Proposal: Real Time Volumetric Rendering of Fog/Smoke

For our CSEE147 project, we propose a real time volumetric rendering of fog/smoke/clouds that can render at an optimized 30+ frames per second on the GPU. We will implement simulated light scattering, light extinction, shadow rays, animated fluid simulation and colored lighting.

Required Libraries/API/Frameworks:

1. GLFW, GLAD, GLM
 - a. Can all be installed with the vcpkg package manager
2. OpenGL
3. OptiX SDK 8.x

Planned Timeframe:

1. For the initial phase of our project, we will focus on developing volume generation through a 3D noise kernel, generating
 - a. This phase will also include general project set up, which includes setting up a 3D texture pipeline and a GLM view
2. We will then implement the Beer lambert extinction and the Henyey greenstein phase function for directional scattering and light extinction.
3. For the third phase, we will focus on expanding our lighting system to one that supports multiple lights with distinct colors.
4. For the final phase, we will implement the fluid animation through an advection kernel using semi lagrangian advection

Potential risks or technical problems/challenges:

1. Warp Divergence
 - a. Optimization concerning frame rate is important to volumetric rendering; however, due to the complexity of the project, warp divergence, which directly affects the framerate, will become an issue. Therefore, a large focus for this project will be avoiding Warp Divergence through general coding practice but also through overall optimized kernel implementation.
2. Shadow rays can be computationally expensive, which would directly affect the frame rate of our project
 - a. This concern raises a dilemma with the overall approach to our project. A preference towards detail and design or towards better performance. However, reducing the quality of the shadow rays by skipping the number

of rays that need to be computed could result in a middle ground solution with both quality and performance.

Minimum Deliverables:

We will deliver at minimum a volumetric rendering of fog/smoke/clouds at an optimized 30+ frames with light scattering and light extinction.